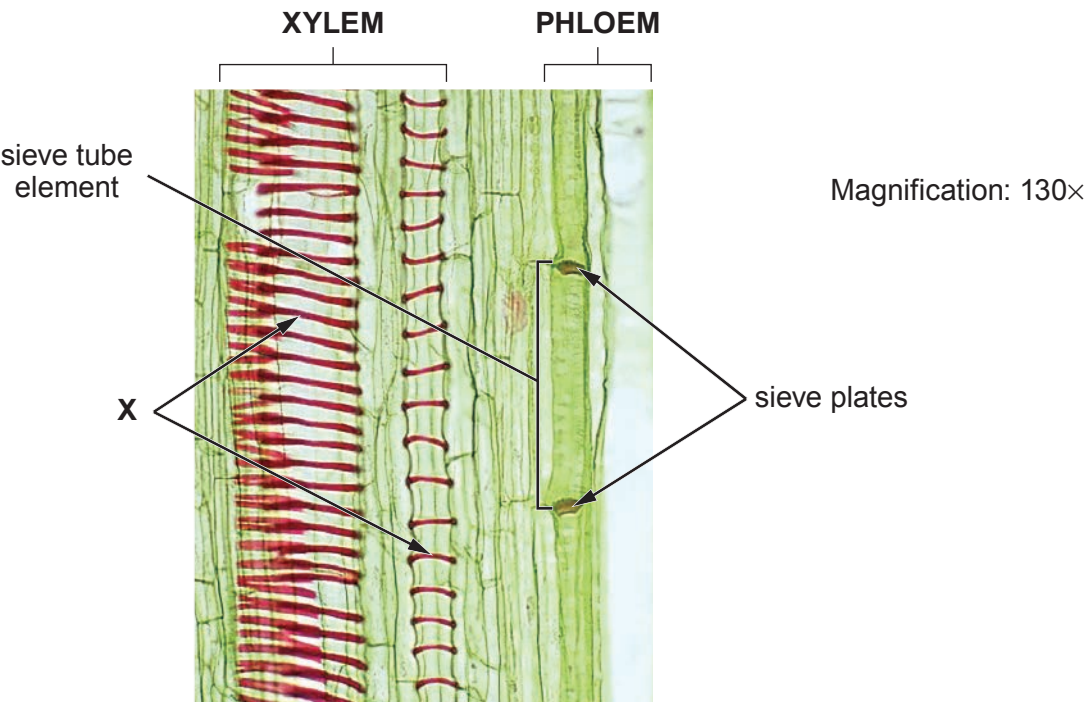


5. **Image 5.1** shows a longitudinal section through a vascular bundle of a marrow (*Cucurbita*) stem.

Image 5.1



- (a) (i) State the main chemical found in the structures labelled **X** in **Image 5.1** and explain the importance of these structures in the functioning of xylem. [2]

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- (ii) Calculate the length of the sieve tube element shown in **Image 5.1** in micrometres (μm). Show your working. [2]

Length = μm

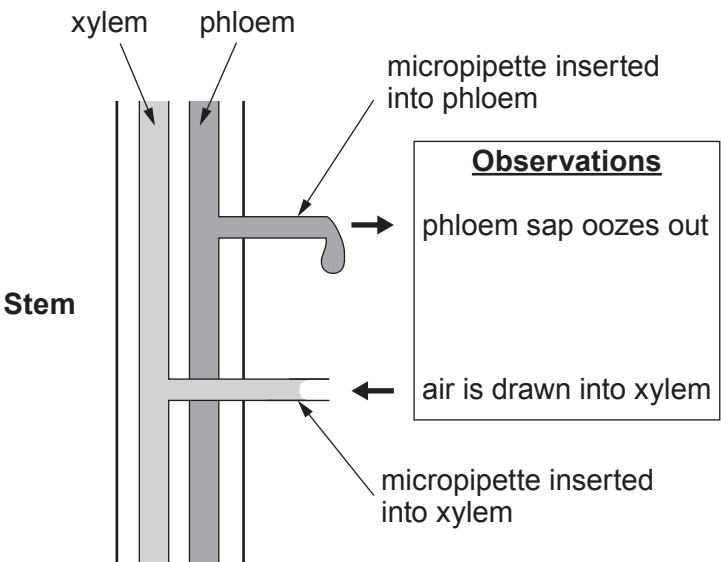
- (iii) Use your answer to (ii) to calculate the length of time, in seconds, that it would take for a molecule of sucrose to travel between the sieve plates labelled in **Image 5.1**, if the flow rate of phloem contents was 0.28 mm s^{-1} . Show your working. [3]

Time = seconds



- (b) In an investigation into the mechanisms of transport, fluid filled micropipettes were inserted into xylem vessels and sieve tube elements. **Image 5.2** shows the observations made.

Image 5.2



Using your knowledge of plant transport, explain the observations shown in **Image 5.2**. [4]

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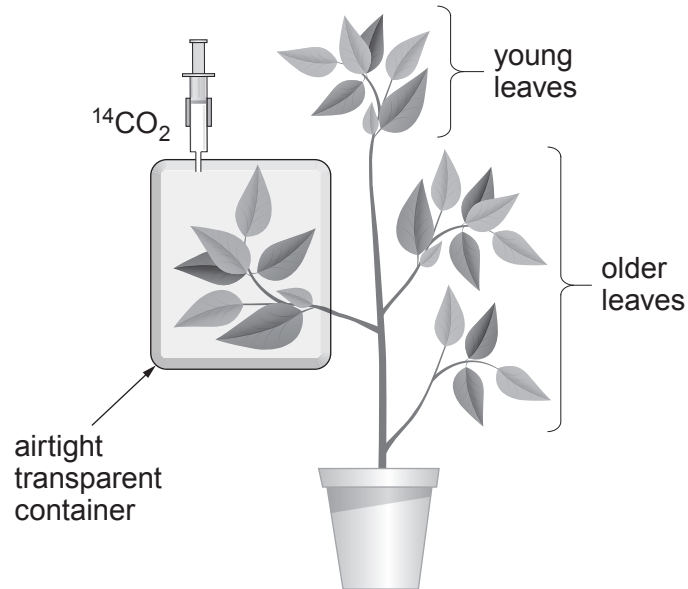
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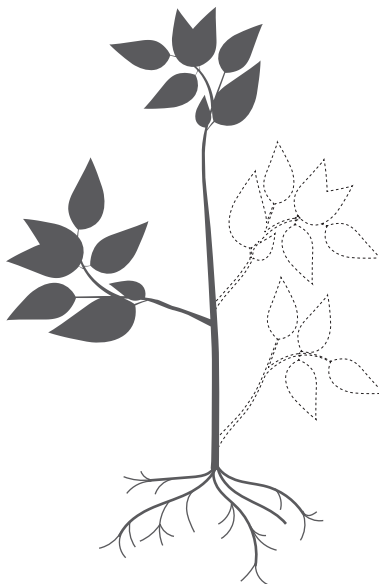
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- (c) In an experiment, some leaves of a plant were supplied with radioactively labelled carbon dioxide ($^{14}\text{CO}_2$) and allowed to photosynthesise, as shown in **Image 5.3**. The leaves used $^{14}\text{CO}_2$ to produce sucrose. The sucrose was then transported from the leaves by the phloem.

Image 5.3

After 24 hours exposure to $^{14}\text{CO}_2$ the plant was removed from the pot, dried, and then laid on photographic film in the dark. Radioactivity exposes the film turning it black. **Image 5.4** represents an autoradiograph showing the location of compounds containing ^{14}C .

Image 5.4

- (i) Other than sucrose, name **one** type of compound containing ^{14}C that may be transported in phloem.

[1]



- (ii) State the conclusions that can be drawn from the distribution of radioactivity in the **leaves** of the plant. [4]

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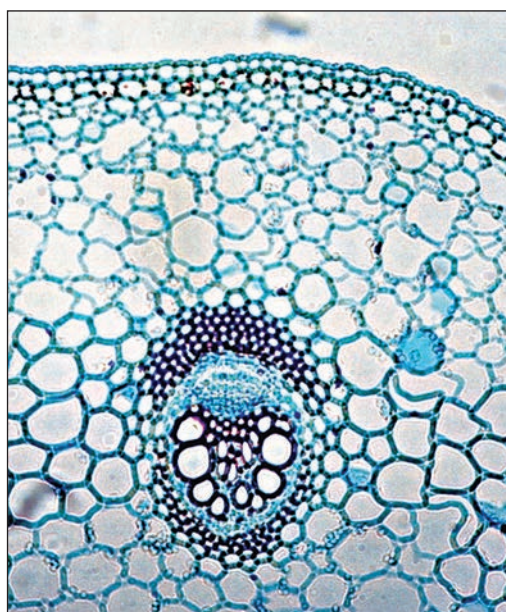
- (iii) The autoradiograph shows that radioactivity is also found in the roots. State the additional information that this provides about transport in phloem. [1]

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- (iv) **Image 5.5** shows a transverse section of part of the stem of this plant.

Image 5.5



Using a single arrow, indicate **one** region on **Image 5.5** in which radioactivity would be found. [1]

